PHYSICS WATER COOLER

Ancient Atmospheres

From a Bombarded Ball of Rock to the Air We Breathe

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Ancient Atmospheres

This episode follows Earth's atmosphere on its 4.5-billion-year journey. It begins with the birth of the solar system and the violent early Earth, including the collision that made the Moon. It then traces Earth's three atmospheres: the first, of hydrogen and helium, soon lost; the second, built from volcanic gases with essentially no free oxygen; and the third, oxygen-rich atmosphere we breathe today. Along the way, life appears, cyanobacteria invent oxygen-producing photosynthesis, and the oxygen in the third atmosphere rises and falls through six stages, bringing Snowball Earths, the Boring Billion, the Cambrian Explosion, and giant insects.

Your Host

Name	Role
Robert Mahoney	Host; physics and astronomy educator

Learning Objectives

After this episode you should be able to:

- Describe how the solar system formed from a nebula, and how a supernova may have helped trigger its collapse.
- Describe the early Earth of the Hadean Eon, including the heavy bombardment and the Moon-forming collision with Theia.
- Name Earth's three atmospheres, describe the composition of each, and explain why the first one was lost.
- Explain how Earth's oceans formed, and identify three possible sources of Earth's water.
- Describe the earliest life on Earth, and explain why the lack of an ozone layer mattered.
- Explain how cyanobacteria and oxygenic photosynthesis changed Earth, and why the first oxygen never reached the air.
- Describe the Great Oxygenation Event and explain how it led to the first Snowball Earth.
- Name and describe the six stages of oxygen in Earth's third atmosphere, with their approximate dates.
- Explain the balance between photosynthesis and respiration, and describe the composition of today's atmosphere.

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QUICK REFERENCE

Key Terms

Term	Definition
Supernova	The explosive death of a massive star; its shockwave may have helped trigger the collapse of the cloud that formed our solar system.
Nebula	A vast cloud of gas and dust in space.
Hadean Eon	Earth's earliest eon, from its formation about 4.54 billion years ago, when the planet was hell-like and heavily bombarded.
Planetesimal	One of the small bodies of rock and ice that collided and merged to build the planets.
Heavy bombardment	The early period, lasting hundreds of millions of years, when space rocks pounded the Earth almost continuously.
Theia	The Mars-sized protoplanet whose side-swiping collision with Earth created the Moon.
Tectonic plates	The moving pieces of Earth's crust; today there are roughly seven major and eight minor plates.
First (primordial) atmosphere	Earth's original atmosphere, mostly hydrogen and helium, lost to space early on.
Solar wind	The stream of charged particles flowing out from the Sun; the fierce solar wind of the young Sun helped strip away Earth's first atmosphere.
Second atmosphere	Earth's atmosphere built from volcanic gases: mainly carbon dioxide and nitrogen, with essentially no free oxygen.
Third atmosphere	The oxygen-rich atmosphere we breathe today; it began with the Great Oxygenation Event.
Free oxygen	Oxygen gas (O_2) that is not chemically bound up in rocks, water, or other compounds.
Archean Eon	The eon that followed the Hadean, during which life first appeared and cyanobacteria evolved.
Anaerobic	Living without oxygen, as Earth's earliest microbes did.
Archaea	A group of single-celled microbes; some early archaea consumed carbon dioxide and released methane as a waste product.
Ozone layer	The layer of ozone high in the atmosphere that shields living things from the Sun's ultraviolet radiation.
Cyanobacteria	Bacteria that evolved oxygenic photosynthesis and filled Earth's oceans and air with oxygen.
Oxygenic photosynthesis	Using sunlight to turn carbon dioxide and water into food, releasing oxygen.

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Term	Definition
Banded iron formations	Rusty, layered rock formed when early oxygen reacted with iron dissolved in the oceans; still mined today.
Proterozoic Eon	The eon that followed the Archean, beginning about 2.5 billion years ago, during which oxygen first entered the air.
Great Oxygenation Event	The rise of oxygen in the air about 2.4 billion years ago; it marks the start of the third atmosphere (Stage 1).
Greenhouse gas	A gas, such as methane or carbon dioxide, that traps heat and warms the planet.
Snowball Earth	A time when Earth froze over, from the poles to near or at the equator.
Boring Billion	The long, uneventful stretch of low, steady oxygen that followed the first rise (Stage 2).
Supercontinent	A single landmass made of most or all of the continents; the breakup of one helped drive the Snowball Earths of about 700 million years ago.
Phanerozoic Eon	The current eon, beginning about 540 million years ago, the age of abundant complex life.
Cambrian Explosion	The rapid diversification of complex animal life, about 540 million years ago, when most major animal body plans appear in the fossil record (Stage 4).
Respiration	Taking in oxygen and giving off carbon dioxide; all living things do it, around the clock.
Parts per million (ppm)	A way of expressing very small fractions; today's carbon dioxide is about 425 parts per million, or 0.04 percent.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
4.6 billion years ago	A little before this, a nearby supernova helped trigger the collapse of the solar nebula
4.54 billion years ago	Earth formed
4.51 billion years ago	The Moon formed, after the collision with Theia
A few cm per year	How fast the Moon drifts away from Earth
384,000 km (239,000 mi)	The Moon's present average distance from Earth
7 and 8	Today's major and minor tectonic plates
3	Earth's atmospheres: first, second, and third
6	Stages of oxygen within the third atmosphere

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Value	What it refers to
~27 atmospheres	One estimate of the pressure when Earth's surface first cooled below the boiling point of water
Millions of years	How long it rained, off and on, to fill the oceans
More than 70%	Share of Earth's surface the oceans eventually covered
3.8–3.5 billion years ago	When the first single-celled life appeared
4.1–4.2 billion years ago	Debatable hints that life began even earlier
3.4–2.9 billion years ago	When cyanobacteria evolved oxygenic photosynthesis, according to recent genetic studies
2.4 billion years ago	The Great Oxygenation Event; the third atmosphere begins (Stage 1)
Most of a billion years	Length of the Boring Billion (Stage 2)
800 million years ago	Oxygen starts climbing again (Stage 3)
700 million years ago	The second round of Snowball Earths
~10%; 540 million years ago	Atmospheric oxygen at the Cambrian Explosion (Stage 4)
470 million years ago	Plants begin greening the continents
~35%; 290 million years ago	The oxygen peak, the age of giant insects (Stage 5)
~21%	Oxygen today (Stage 6)
~78%	Nitrogen today
0.04% (~425 ppm)	Carbon dioxide today

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QUICK REFERENCE

The Six Stages of the Third Atmosphere

Stage	What happened
Stage 1 — about 2.4 billion years ago	The Great Oxygenation Event: oxygen enters the air, the first ozone layer forms, and the first Snowball Earth follows.
Stage 2 — the Boring Billion	Oxygen falls back and holds low and steady for most of a billion years.
Stage 3 — from about 800 million years ago	Oxygen climbs again, the ozone layer thickens, and conditions are set for complex life; more Snowball Earths about 700 million years ago.
Stage 4 — about 540 million years ago	Oxygen reaches about 10%; the Cambrian Explosion.
Stage 5 — about 290 million years ago	Oxygen peaks near 35%; the age of giant insects.
Stage 6 — today	Oxygen declines and settles near 21%: the air we breathe right now.

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — The Genesis of the Earth and Moon (The Hadean Eon)

A little over **4.6 billion years ago**, a nearby star ended its life in a **supernova**. Its shockwave helped trigger the collapse of a vast cloud of gas and dust, a **nebula**, and seeded it with heavier elements forged in the dying star. The cloud flattened into a spinning disk with a central bulge that became our **Sun**.

Earth formed about 4.54 billion years ago. Its earliest state was hell-like, almost continuously bombarded by meteoroids, asteroids, comets, and **planetesimals** for hundreds of millions of years. Early on, a Mars-sized protoplanet named **Theia** side-swiped Earth. The debris first formed a ring around Earth, then gathered into the **Moon**, about **4.51 billion years ago**. The Moon began very close to Earth and has been drifting outward a few centimeters a year ever since, to its present average distance of about **384,000 kilometers** (239,000 miles).

Earth's cooling crust would much later fracture into today's moving **tectonic plates**, roughly seven major and eight minor ones, though modern plate tectonics took a long time to get going. Throughout, countless volcanoes spewed gases into the sky.

Earth has had **three atmospheres**. The **first**, or primordial, atmosphere was mostly **hydrogen and helium**, and it escaped: the gases were too light for gravity to hold, or were stripped away by the fierce **solar wind** of the young Sun. A brief atmosphere of vaporized rock followed the Moon-forming impact. What replaced them was the **second atmosphere**, built from **volcanic gases**: mainly **carbon dioxide and nitrogen**, with water vapor, sulfur dioxide, and other gases, and **essentially no free oxygen**. The oxygen-rich **third atmosphere** we breathe today has been anything but stable: its oxygen has risen and fallen through **six stages**.

When Earth's surface finally cooled below the boiling point of water, under a pressure of perhaps **27 atmospheres**, it began to **rain**, off and on, for millions of years. The water was partly native to Earth, possibly partly delivered by Theia, and partly brought later by asteroids and comets. The oceans filled, eventually covering **more than 70%** of the planet.

Part 2 — The Beginning of Life on Earth (The Archean Eon)

The first **single-celled life** appeared sometime between roughly **3.8 and 3.5 billion years ago**, possibly near hot, mineral-rich cracks on the ocean floor. Some researchers see debatable hints of life as early as 4.1 to 4.2 billion years ago.

The air held almost no free oxygen, and therefore **no ozone**, so the Sun's DNA-shredding **ultraviolet** radiation poured onto the surface unchecked. The earliest microbes were **anaerobic**. Some, the **archaea**, consumed the then-abundant carbon dioxide dissolved in water and released **methane** as a waste product.

Sometime in the Archean, likely between about **3.4 and 2.9 billion years ago** according to recent genetic studies, **cyanobacteria** evolved **oxygenic photosynthesis**: using sunlight, they consumed carbon dioxide and water and released oxygen. At first, that oxygen **never reached**

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the air. It reacted with iron dissolved in the oceans, forming rusty layers on the ocean floor, the **banded iron formations** we still mine today. Microbes nearest the surface were killed by ultraviolet light, but layers beneath were shielded by the dead cells above, so life persisted. All this still happened under the second atmosphere.

Part 3 — The Third Atmosphere Begins: Stages 1 Through 3 (The Proterozoic Eon)

By about **2.4 billion years ago**, the oceans' dissolved iron was largely used up, and oxygen began escaping into the air. High up, it formed the **first true ozone layer**. This is the **Great Oxygenation Event**, the arrival of the third atmosphere: **Stage 1**.

Methane had been an important **greenhouse gas**, keeping Earth warm under the young, fainter Sun. Oxygen destroyed that methane, and the planet plunged into a deep freeze, a **Snowball Earth**. Falling carbon dioxide also played a role.

After the first surge, oxygen fell back and held low and steady for most of a billion years, the **Boring Billion**: **Stage 2**. Then, beginning around **800 million years ago**, oxygen started climbing again, a thicker ozone layer formed, and conditions were set for complex life: **Stage 3**. Plants would not green the continents until about **470 million years ago**, with animals following.

Around **700 million years ago**, Earth froze over again in another round of **Snowball Earths**, driven mainly by the breakup of a **supercontinent**, intense rock weathering, and the resulting drawdown of carbon dioxide.

Part 4 — Stages 4, 5, and 6 (The Phanerozoic Eon)

When oxygen climbed to around **10%**, about **540 million years ago**, complex animal life diversified with astonishing speed in the **Cambrian Explosion**, establishing the lineages of most animal life today: **Stage 4**. Simple multicellular life had appeared tens of millions of years earlier, but most major animal body plans first appear in the fossil record in the Cambrian.

Oxygen kept climbing, fitfully, until about **290 million years ago**, when by the best model estimates it peaked near **35%**: the age of **giant insects**, **Stage 5**. Oxygen then declined, for reasons including vast wildfires, settling near today's **21%**: **Stage 6**, the air we breathe now.

Part 5 — The Current Atmosphere, Current Life, and a Forward-Looking Perspective

During the day, plants, algae, and cyanobacteria **photosynthesize**, taking in carbon dioxide and releasing oxygen. But all living things also **respire**, taking in oxygen and giving off carbon dioxide, around the clock. In daylight, plants photosynthesize faster than they respire, so they release oxygen on balance; at night, only respiration remains.

Carbon dioxide, once the dominant gas, is now about **0.04%** of the atmosphere, roughly **425 parts per million**. Nitrogen and oxygen make up almost all the rest, at about **78%** and **21%**. Many atmospheric scientists are concerned that rising carbon dioxide, driven largely by burning fossil fuels, is overheating the Earth, bringing rising seas, disrupted jet streams and ocean currents, and more extreme weather.

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From a hellish, bombarded ball of rock to the only known planet supporting life, Earth's atmosphere has traveled an extraordinary 4.5-billion-year journey through **three great atmospheres**, with **six stages** within the third.

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STUDY QUESTIONS

Questions with Model Answers

Question numbering matches the student study guide exactly. Model answers are written at the level expected of a strong student response, not as the only acceptable wording.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — The Genesis of the Earth and Moon (The Hadean Eon)

1. What event, a little over 4.6 billion years ago, may have helped start the formation of our solar system? What two things did it do to the nebula?

ANSWER A nearby star ended its life in a supernova. Its shockwave helped trigger the collapse of the nebula, and it seeded the cloud with heavier elements forged in the dying star.

2. How did the nebula's shape change as it collapsed? What became of the bulge at its center?

ANSWER It was haphazard at first, then flattened out and began to spin, forming a disk of gas and solids with a bulge at the center. The bulge became our Sun.

3. When did Earth form? Describe its earliest state.

ANSWER About 4.54 billion years ago. Its earliest state was hell-like, almost continuously bombarded by meteoroids, asteroids, comets, and planetesimals for hundreds of millions of years.

4. How and when did the Moon form?

ANSWER A Mars-sized protoplanet named Theia side-swiped Earth. The Moon coalesced out of the debris ejected by the impact, first forming a ring around Earth and then gathering into a sphere, about 4.51 billion years ago.

5. Where did the Moon begin its life, and how has its distance from Earth changed? What is its average distance today?

ANSWER It began very close to Earth and has been drifting outward a few centimeters a year ever since. Its present average distance is about 384,000 kilometers, or 239,000 miles.

6. What did Earth's cooling crust eventually become? How many major and minor pieces are there today?

ANSWER Much later, it fractured into the moving tectonic plates we know today: roughly seven major and eight minor plates.

7. How many atmospheres has Earth had? Name them.

ANSWER Three: the first (primordial) atmosphere, the second atmosphere, and the oxygen-rich third atmosphere we breathe today.

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8. What was Earth's first atmosphere made of, and what are the two reasons it was lost?

ANSWER Mostly hydrogen and helium. These gases were too light for Earth's gravity to hold onto, or were stripped away by the fierce solar wind of the young Sun.

9. Where did Earth's second atmosphere come from? What was it mostly made of, and what crucial gas was missing?

ANSWER It was built from volcanic gases: primarily carbon dioxide and nitrogen, with water vapor, sulfur dioxide, and other gases. There was essentially no free oxygen.

10. When did it begin to rain on Earth, and for how long? Roughly what pressure was the atmosphere under?

ANSWER When Earth's surface temperature finally dropped below the boiling point of water. It rained, off and on, for millions of years. By one estimate, the pressure was around 27 atmospheres.

11. Name three possible sources of Earth's water. How much of Earth did the oceans eventually cover?

ANSWER Water native to Earth, locked in the rocks it formed from; water possibly delivered by Theia, which may have arrived rich in water; and water brought later by asteroids and comets. The oceans eventually covered more than 70% of Earth.

Part 2 — The Beginning of Life on Earth (The Archean Eon)

1. When and where did the first single-celled life probably appear? What earlier date has been suggested?

ANSWER Sometime between roughly 3.8 and 3.5 billion years ago, possibly near hot, mineral-rich cracks on the ocean floor. Some researchers have found hints of life as early as 4.1 to 4.2 billion years ago, though this remains debatable.

2. Why was there no ozone layer during the Archean Eon? What did that mean for life on the surface?

ANSWER The air held almost no free oxygen, and ozone is made from oxygen. Without it, the Sun's DNA-shredding ultraviolet radiation poured down onto the surface unchecked.

3. What does “anaerobic” mean? How did the archaea make a living, and what waste gas did they produce?

ANSWER Living without oxygen. Some archaea made a living through chemical reactions that consumed the then-abundant carbon dioxide dissolved in water, releasing methane as a waste product.

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4. What "trick" did cyanobacteria evolve, and roughly when? What does the process take in and give off?

ANSWER Oxygenic photosynthesis, sometime in the Archean Eon, about 3.4 to 2.9 billion years ago according to recent genetic studies. Using sunlight, it takes in carbon dioxide and water and releases oxygen.

5. Why did the first oxygen made by cyanobacteria never reach the air? What evidence of it can we still see today?

ANSWER It reacted immediately with iron dissolved in the early oceans, forming rusty mineral layers that settled to the ocean floor. These are the banded iron formations, which we still mine today.

6. How did life survive the ultraviolet radiation near the ocean surface?

ANSWER The microbes nearest the surface were killed, but the layers just beneath them were shielded by the dead cells above, so life persisted and kept pumping out oxygen.

7. Under which of Earth's atmospheres did all of the events of the Archean Eon unfold?

ANSWER The second atmosphere. The third atmosphere, the oxygen age, had not yet begun.

Part 3 — The Third Atmosphere Begins: Stages 1 Through 3 (The Proterozoic Eon)

1. What happened about 2.4 billion years ago that let oxygen finally enter the air? What formed high in the atmosphere?

ANSWER The oceans' dissolved iron was largely used up, so oxygen began escaping into the air. High in the atmosphere, it formed the first true ozone layer.

2. What is the Great Oxygenation Event? Which atmosphere and which stage does it begin?

ANSWER The turning point, about 2.4 billion years ago, when oxygen began entering the air. It marks the arrival of Earth's third atmosphere, the oxygen age, and is Stage 1.

3. Why did the rise of oxygen plunge Earth into a Snowball Earth? What other gas played a role?

ANSWER Methane had been an important greenhouse gas keeping Earth warm under the young, fainter Sun. Oxygen destroyed the methane, and with that greenhouse blanket gone, the planet froze. Falling carbon dioxide also played a role.

4. What is the Boring Billion? Which stage of the third atmosphere is it?

ANSWER After the initial surge, oxygen fell back and held low and steady for the better part of a billion years, a stretch so uneventful that geologists half-jokingly call it the Boring Billion. It is Stage 2.

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5. What began around 800 million years ago? Which stage is it?

ANSWER Oxygen levels started climbing again, a thicker ozone layer formed, and conditions were set for complex life. This is Stage 3.

6. When did plants begin greening the continents?

ANSWER Around 470 million years ago, with animals following.

7. When did the second round of Snowball Earths occur, and what three things drove it?

ANSWER Around 700 million years ago. It was driven mainly by the breakup of a supercontinent, intense rock weathering, and the carbon dioxide drawdown that came with it.

Part 4 — Stages 4, 5, and 6 (The Phanerozoic Eon)

1. What was the Cambrian Explosion? When did it happen, and about how much oxygen was in the air?

ANSWER The rapid diversification of complex animal life, establishing the lineages of most current animal life forms. It happened about 540 million years ago, when oxygen had climbed to around 10%. It is Stage 4.

2. Multicellular life existed before the Cambrian. So what is special about the Cambrian?

ANSWER Simple multicellular life had appeared tens of millions of years earlier, but the Cambrian is when most of the major animal body plans we know today first burst into the fossil record.

3. When did oxygen reach its peak, and how high did it get? What kind of animals is this period known for?

ANSWER About 290 million years ago, near 35% by the best model estimates. It was the age of giant insects: Stage 5.

4. What happened to oxygen after its peak? Give one reason, and today's oxygen level.

ANSWER It declined again, for a variety of reasons including vast wildfires, settling near today's level of about 21%. This is Stage 6, the air we breathe right now.

5. List the six stages of oxygen in the third atmosphere, with an approximate date and one key event for each.

ANSWER Stage 1, about 2.4 billion years ago: the Great Oxygenation Event and first ozone layer. Stage 2: the Boring Billion. Stage 3, from about 800 million years ago: oxygen climbs again. Stage 4, about 540 million years ago: about 10% oxygen and the Cambrian Explosion. Stage 5, about 290 million years ago: the 35% peak and giant insects. Stage 6, today: about 21%.

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Part 5 — The Current Atmosphere, Current Life, and a Forward-Looking Perspective

1. What do plants, algae, and cyanobacteria do during the day through photosynthesis?

ANSWER They pull carbon dioxide from the air and release oxygen.

2. What is respiration, and which living things do it? When?

ANSWER Taking in oxygen and giving off carbon dioxide. All living things do it, including animals, plants, fungi, insects, and microbes, constantly, around the clock.

3. Why do plants release oxygen on balance during the day, but not at night?

ANSWER In daylight they photosynthesize faster than they respire. At night, photosynthesis switches off and only respiration remains.

4. How much of today's atmosphere is carbon dioxide? Express it both as a percentage and in parts per million.

ANSWER About 0.04%, or roughly 425 parts per million: a very small but crucial fraction, once the dominant gas.

5. What two gases make up almost all the rest of today's atmosphere, and in what proportions?

ANSWER Nitrogen, about 78%, and oxygen, about 21%.

6. Why are many atmospheric scientists concerned about rising carbon dioxide? Name three consequences mentioned.

ANSWER Rising carbon dioxide, driven largely by burning fossil fuels, is overheating the Earth. Consequences include rising sea levels, disruptions in the jet streams and ocean currents, and more extreme weather.

7. Summarize the journey of Earth's atmosphere in one or two sentences, naming its atmospheres and stages.

ANSWER From a hellish, bombarded ball of rock to the only known planet supporting life, Earth's atmosphere evolved over 4.5 billion years through three great atmospheres, with six stages of oxygen within the third, and its journey continues.

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TEACHER NOTES

Corrections and Cautions

Points in the episode that need a qualification, or a prepared answer for the question a sharp student will ask.

C1 **A paragraph is missing from the recording.**

The script's passage about the crust forming, iron sinking to build Earth's core, and the core generating the magnetic field is not in the video, so Part 1 jumps from the Moon's distance to “That cooling crust…”. These guides follow the video. If you restore the passage, add the core and magnetic field to Part 1.

C2 **The supernova trigger is a hypothesis.**

Short-lived isotopes such as aluminum-26 in meteorites show that material from a massive star reached the young solar system, but whether a supernova shockwave actually triggered the collapse is still debated; winds from a massive Wolf-Rayet star are one alternative.

C3 **The Moon's age is uncertain.**

4.51 billion years is at the old end of the estimates; others range down to about 4.35 billion. The Moon now recedes about 3.8 centimeters a year, measured by bouncing lasers off reflectors left by Apollo astronauts.

C4 **The boiling point under 27 atmospheres.**

At 27 atmospheres, water boils at about 230°C (446°F), so the first rain fell on a surface far hotter than today's boiling point. The 27-atmosphere figure is one model estimate.

C5 **The sources of water are still debated.**

Hydrogen isotope ratios suggest that most of Earth's water came from rocky, asteroid-like material rather than comets. The idea that Theia was water-rich comes from a few studies and is not settled.

C6 **Early oceans, early continents.**

“More than 70%” is today's figure (about 71%). The early Earth had little continental crust, and some scientists think it may have been almost entirely covered by water.

C7 **What the archaea ate.**

Methanogens combine carbon dioxide with hydrogen (from volcanic and hydrothermal sources) to make methane. The narration simplifies this to “consumed carbon dioxide.”

C8 **Dating oxygenic photosynthesis.**

The 3.4 to 2.9 billion year range comes from molecular-clock studies; chemical “whiffs” of oxygen in rocks about 2.5 to 3.0 billion years old support an origin well before the Great Oxygenation Event. The original script said “well before 2.7 billion years ago.”

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C9 **Shielding from ultraviolet light.**

Water itself also blocked ultraviolet radiation; a few meters of water protected microbes living below the surface. Microbial mats, like the stromatolites still found in Australia, layered living cells under dead ones.

C10 **Oxygen had more than iron to fight.**

Volcanic gases and other “oxygen sinks” also consumed early oxygen. The Great Oxygenation Event is usually dated about 2.4 to 2.3 billion years ago, and oxygen in the air was still well under 1% of today's level at first.

C11 **The first Snowball Earth.**

This is the Huronian glaciation, about 2.4 to 2.1 billion years ago. Whether it froze the whole planet is less certain than for the later Snowball Earths. The young Sun was only about 70 to 75% as bright as today's, which is why the greenhouse blanket mattered so much.

C12 **The Part 3 title in the video.**

The recording titles Part 3 “The Boring Billion Plus Years” and Part 4 “Multicellular Life in the Cambrian Explosion,” while the script titles them by stage. The guides use the script's titles.

C13 **The later Snowball Earths.**

The Sturtian (about 717 to 660 million years ago) and Marinoan (about 650 to 635 million years ago) glaciations. The supercontinent that broke up was Rodinia.

C14 **Oxygen percentages are model estimates.**

The 10% at the Cambrian and the 35% peak come from geochemical models; some newer models put the peak nearer 25 to 30%. Giant insects such as the griffinfly Meganeura, with a wingspan of about 70 centimeters (2.3 feet), lived in the late Carboniferous and early Permian.

C15 **Why oxygen declined.**

The main drivers were changes in how much organic carbon was buried and a drier Permian climate. Wildfires act mainly as a brake: in air richer in oxygen, fires ignited by lightning burn far more easily, which limits how high oxygen can go.

C16 **Carbon dioxide keeps rising.**

About 425 parts per million was the 2025 average, up from about 280 before the Industrial Revolution and still rising by roughly 2 to 3 parts per million a year.

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TEACHER NOTES

Teaching the Episode

Build the calendar.

Run the one-year calendar from the Try It box on a long strip of paper on the wall. Students are usually surprised that the air had no free oxygen until late June, and that complex animals only arrive in mid-November.

Draw the oxygen curve.

Have students sketch oxygen against time, marking the six stages. The rise, the long flat Boring Billion, the climb, the 35% peak, and the fall to 21% make the stages easy to remember.

Rust in a jar.

Steel wool in a sealed jar of water slowly rusts and uses up the oxygen. Connect it to the iron in the early oceans soaking up the first oxygen and forming banded iron formations.

Show a banded iron sample.

A polished piece of banded iron formation (or a photo) shows the alternating rust-red and gray bands, a physical record of oxygen made by cyanobacteria billions of years ago.

Photosynthesis versus respiration.

Bromothymol blue changes color with carbon dioxide. A water plant in the indicator turns it blue in light and back toward yellow in the dark, showing the day-and-night balance from Part 5.

Keep the bookkeeping straight.

Students often mix up atmospheres and stages. Stress that there are three atmospheres, and that the six stages all belong to the third, oxygen-rich one.

Connect to the atmosphere module.

Pair this episode with Module 5A on today's atmosphere: the ozone layer that first formed at the Great Oxygenation Event is the same shield in today's stratosphere.

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